

# Name: Bassant Ali Kamal Ali

## Track: Cloud Platform Development

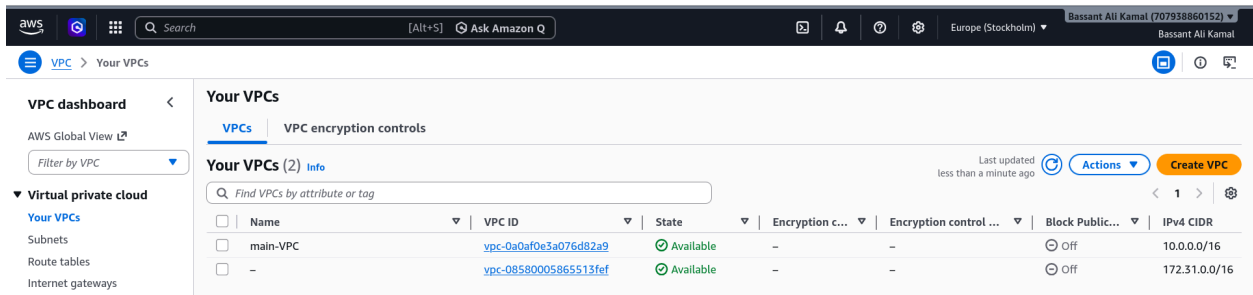
## Course: Terraform

## Lab 1

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## Networking

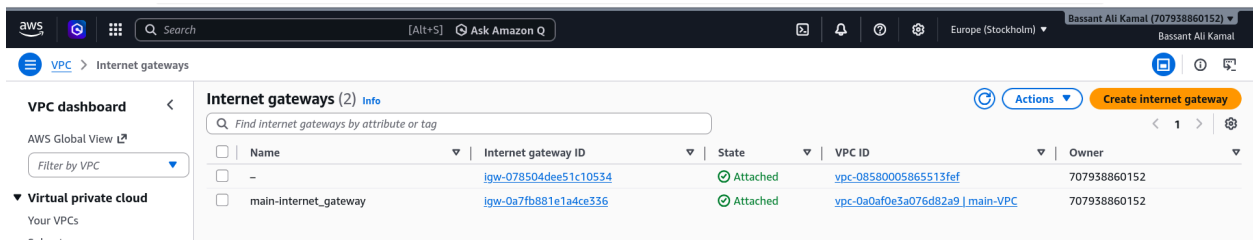
### 1- create vpc



The screenshot shows the AWS Management Console interface for VPCs. The left sidebar contains navigation links for VPC dashboard, Virtual private cloud, Subnets, Route tables, and Internet gateways. The main content area displays 'Your VPCs (2)' with a table listing two VPCs: 'main-VPC' and an unnamed VPC. Both are in an 'Available' state.

| Name     | VPC ID                | State     | Encryption c... | Encryption control ... | Block Public... | IPv4 CIDR     |
|----------|-----------------------|-----------|-----------------|------------------------|-----------------|---------------|
| main-VPC | vpc-0a0af0e3a076d82a9 | Available | -               | -                      | Off             | 10.0.0.0/16   |
| -        | vpc-08580005865513fef | Available | -               | -                      | Off             | 172.31.0.0/16 |

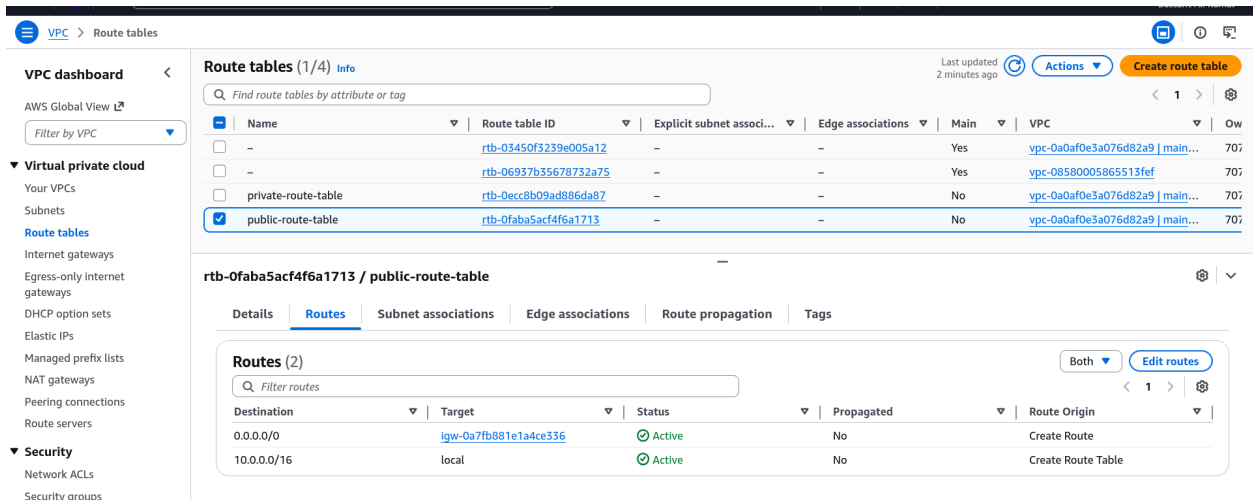
### 2- create internet gateway



The screenshot shows the AWS Management Console interface for Internet gateways. The left sidebar contains navigation links for VPC dashboard, Virtual private cloud, Subnets, Route tables, and Internet gateways. The main content area displays 'Internet gateways (2)' with a table listing two gateways: 'main-Internet\_gateway' and an unnamed gateway. Both are in an 'Attached' state.

| Name                  | Internet gateway ID   | State    | VPC ID                           | Owner        |
|-----------------------|-----------------------|----------|----------------------------------|--------------|
| -                     | igw-078504dee51c10534 | Attached | vpc-08580005865513fef            | 707938860152 |
| main-Internet_gateway | igw-0a7fb881e1a4ce336 | Attached | vpc-0a0af0e3a076d82a9   main-VPC | 707938860152 |

### 3- create public route table



The screenshot shows the AWS Management Console interface for Route tables. The left sidebar contains navigation links for VPC dashboard, Virtual private cloud, Subnets, Route tables, Internet gateways, Egress-only internet gateways, DHCP option sets, Elastic IPs, Managed prefix lists, NAT gateways, Peering connections, Route servers, Security, Network ACLs, and Security groups. The main content area displays 'Route tables (1/4)' with a table listing four route tables: 'rtb-03450f3239e005a12', 'rtb-06937b35678732a75', 'private-route-table', and 'public-route-table'. The 'public-route-table' is selected. Below the table, the 'Routes' section shows two routes: '0.0.0.0/0' and '10.0.0.0/16', both in an 'Active' state.

| Name                | Route table ID        | Explicit subnet associ... | Edge associations | Main | VPC                             | Owner  |
|---------------------|-----------------------|---------------------------|-------------------|------|---------------------------------|--------|
| -                   | rtb-03450f3239e005a12 | -                         | -                 | Yes  | vpc-0a0af0e3a076d82a9   main... | 707... |
| -                   | rtb-06937b35678732a75 | -                         | -                 | Yes  | vpc-08580005865513fef           | 707... |
| private-route-table | rtb-0ecc8b09ad886da87 | -                         | -                 | No   | vpc-0a0af0e3a076d82a9   main... | 707... |
| public-route-table  | rtb-0fab5ac4f6a1713   | -                         | -                 | No   | vpc-0a0af0e3a076d82a9   main... | 707... |

rtb-0fab5ac4f6a1713 / public-route-table

| Destination | Target                | Status | Propagated | Route Origin       |
|-------------|-----------------------|--------|------------|--------------------|
| 0.0.0.0/0   | igw-0a7fb881e1a4ce336 | Active | No         | Create Route       |
| 10.0.0.0/16 | local                 | Active | No         | Create Route Table |

## 4- create private route table

The screenshot shows the AWS VPC console 'Route tables' page. The left sidebar contains navigation links for VPC dashboard, Virtual private cloud, and Security. The main content area shows a list of route tables. The 'private-route-table' is selected, and its details are displayed below the list.

| Name                | Route table ID        | Explicit subnet associ... | Edge associations | Main | VPC                             | Owner |
|---------------------|-----------------------|---------------------------|-------------------|------|---------------------------------|-------|
| -                   | rtb-03450f3239e005a12 | -                         | -                 | Yes  | vpc-0a0af0e3a076d82a9   main... | 707   |
| -                   | rtb-06937b35678732a75 | -                         | -                 | Yes  | vpc-08580005865513fef           | 707   |
| private-route-table | rtb-0ecc8b09ad886da87 | -                         | -                 | No   | vpc-0a0af0e3a076d82a9   main... | 707   |
| public-route-table  | rtb-0fab5ac4f6a1713   | -                         | -                 | No   | vpc-0a0af0e3a076d82a9   main... | 707   |

**rtb-0ecc8b09ad886da87 / private-route-table**

**Details**

Route table ID: [rtb-0ecc8b09ad886da87](#)

VPC: [vpc-0a0af0e3a076d82a9 | main-VPC](#)

Main: No

Owner ID: [707938860152](#)

Explicit subnet associations: -

Edge associations: -

## 5- create public route

The screenshot shows the AWS VPC console 'Route tables' page. The 'public-route-table' is selected, and its details are displayed below the list.

| Name                | Route table ID        | Explicit subnet associ... | Edge associations | Main | VPC                             | Owner |
|---------------------|-----------------------|---------------------------|-------------------|------|---------------------------------|-------|
| private-route-table | rtb-0ecc8b09ad886da87 | -                         | -                 | No   | vpc-0a0af0e3a076d82a9   main... | 707   |
| public-route-table  | rtb-0fab5ac4f6a1713   | -                         | -                 | No   | vpc-0a0af0e3a076d82a9   main... | 707   |

**rtb-0fab5ac4f6a1713 / public-route-table**

**Details**

Route table ID: [rtb-0fab5ac4f6a1713](#)

VPC: [vpc-0a0af0e3a076d82a9 | main-VPC](#)

Main: No

Owner ID: [707938860152](#)

Explicit subnet associations: -

Edge associations: -

## 6- attach public route table to subnets Compute

The screenshot shows the AWS VPC console 'Route tables' page. The 'public-route-table' is selected, and its 'Subnet associations' are displayed below the list.

| Name            | Subnet ID                                | IPv4 CIDR   | IPv6 CIDR |
|-----------------|------------------------------------------|-------------|-----------|
| Public Subnet 2 | <a href="#">subnet-0d2956c66bbba1f7c</a> | 10.0.2.0/24 | -         |
| Public Subnet 1 | <a href="#">subnet-058e73493146a075d</a> | 10.0.1.0/24 | -         |

**rtb-0fab5ac4f6a1713 / public-route-table**

**Subnet associations (2)**

Find subnet association

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

Route tables (1/4)

Find route tables by attribute or tag

|                                     | Name                | Route table ID        | Explicit subnet associ... | Edge associations | Main | VPC                             | Own |
|-------------------------------------|---------------------|-----------------------|---------------------------|-------------------|------|---------------------------------|-----|
| <input type="checkbox"/>            | -                   | rtb-03450f3239e005a12 | -                         | -                 | Yes  | vpc-0a0af0e3a076d82a9   main... | 707 |
| <input type="checkbox"/>            | -                   | rtb-06937b35678732a75 | -                         | -                 | Yes  | vpc-08580005865513fef           | 707 |
| <input checked="" type="checkbox"/> | private-route-table | rtb-0ecc8b09ad886da87 | 2 subnets                 | -                 | No   | vpc-0a0af0e3a076d82a9   main... | 707 |
| <input type="checkbox"/>            | public-route-table  | rtb-0fab5ac4f6a1713   | 2 subnets                 | -                 | No   | vpc-0a0af0e3a076d82a9   main... | 707 |

rtb-0ecc8b09ad886da87 / private-route-table

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (2)

Find subnet association

| Name             | Subnet ID                | IPv4 CIDR   | IPv6 CIDR |
|------------------|--------------------------|-------------|-----------|
| Private Subnet 2 | subnet-08da3c1d883619882 | 10.0.4.0/24 | -         |
| Private Subnet 1 | subnet-05cd9dd7337b46e43 | 10.0.3.0/24 | -         |

# 7- create security group which allow ssh from 0.0.0.0/0

VPC > Security Groups

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

Security Groups (1/4)

Find security groups by attribute or tag

|                                     | Name                   | Security group ID    | Security group name        | VPC ID                | Description                   |
|-------------------------------------|------------------------|----------------------|----------------------------|-----------------------|-------------------------------|
| <input type="checkbox"/>            | application_SSH + p... | sg-07442679122f228f6 | application-security-group | vpc-0a0af0e3a076d82a9 | SSH + port 3000 from VPC onl  |
| <input checked="" type="checkbox"/> | bastion_SSH            | sg-0d34856d81c2b1e71 | bastion-security-group     | vpc-0a0af0e3a076d82a9 | Allow SSH inbound traffic and |
| <input type="checkbox"/>            | -                      | sg-01d36696c6ede956a | default                    | vpc-0a0af0e3a076d82a9 | default VPC security group    |
| <input type="checkbox"/>            | -                      | sg-09404c093e5454557 | default                    | vpc-08580005865513fef | default VPC security group    |

sg-0d34856d81c2b1e71 - bastion-security-group

Details

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Search

|  | Security group rule ID | IP version | Type | Protocol | Port range | Source    |
|--|------------------------|------------|------|----------|------------|-----------|
|  | sg-0d7d7abd943b9cb90   | IPv4       | SSH  | TCP      | 22         | 0.0.0.0/0 |

sg-0d34856d81c2b1e71 - bastion-security-group

Details

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Outbound rules (1)

Search

|  | Security group rule ID | IP version | Type        | Protocol | Port range | Destination |
|--|------------------------|------------|-------------|----------|------------|-------------|
|  | sg-02c5af1a2f110f62f   | IPv4       | All traffic | All      | All        | 0.0.0.0/0   |

## 8- create security group that allow ssh and port 3000 from vpc cidr only

**Security Groups (1/4)** Info Actions Export security groups to CSV Create security group

Find security groups by attribute or tag

|                                     | Name                   | Security group ID                    | Security group name        | VPC ID                                | Description                   |
|-------------------------------------|------------------------|--------------------------------------|----------------------------|---------------------------------------|-------------------------------|
| <input type="checkbox"/>            | -                      | <a href="#">sg-01d36696c6ede956a</a> | default                    | <a href="#">vpc-0a0af0e3a076d82a9</a> | default VPC security group    |
| <input type="checkbox"/>            | -                      | <a href="#">sg-09404c093e5454557</a> | default                    | <a href="#">vpc-08580005865513fef</a> | default VPC security group    |
| <input type="checkbox"/>            | bastion_SSH            | <a href="#">sg-0d34856d81c2b1e71</a> | bastion-security-group     | <a href="#">vpc-0a0af0e3a076d82a9</a> | Allow SSH inbound traffic and |
| <input checked="" type="checkbox"/> | application_SSH + p... | <a href="#">sg-07442679122f228f6</a> | application-security-group | <a href="#">vpc-0a0af0e3a076d82a9</a> | SSH + port 3000 from VPC onl  |

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**sg-07442679122f228f6 - application-security-group** Settings Dropdown

Details **Inbound rules** Outbound rules Sharing VPC associations Related resources - new Tags

**Inbound rules (2)** Manage tags Edit inbound rules

Search

| Security group rule ID | IP version | Type       | Protocol | Port range | Source      | Description |
|------------------------|------------|------------|----------|------------|-------------|-------------|
| sgr-0d728783fc4988d02  | IPv4       | Custom TCP | TCP      | 3000       | 10.0.0.0/16 | -           |
| sgr-005c55b05542c368a  | IPv4       | SSH        | TCP      | 22         | 10.0.0.0/16 | -           |

**Security Groups (1/4)** Info Actions Export security groups to CSV Create security group

Find security groups by attribute or tag

|                                     | Name                   | Security group ID                    | Security group name        | VPC ID                                | Description                   |
|-------------------------------------|------------------------|--------------------------------------|----------------------------|---------------------------------------|-------------------------------|
| <input type="checkbox"/>            | -                      | <a href="#">sg-01d36696c6ede956a</a> | default                    | <a href="#">vpc-0a0af0e3a076d82a9</a> | default VPC security group    |
| <input type="checkbox"/>            | -                      | <a href="#">sg-09404c093e5454557</a> | default                    | <a href="#">vpc-08580005865513fef</a> | default VPC security group    |
| <input type="checkbox"/>            | bastion_SSH            | <a href="#">sg-0d34856d81c2b1e71</a> | bastion-security-group     | <a href="#">vpc-0a0af0e3a076d82a9</a> | Allow SSH inbound traffic and |
| <input checked="" type="checkbox"/> | application_SSH + p... | <a href="#">sg-07442679122f228f6</a> | application-security-group | <a href="#">vpc-0a0af0e3a076d82a9</a> | SSH + port 3000 from VPC onl  |

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**sg-07442679122f228f6 - application-security-group** Settings Dropdown

Details Inbound rules **Outbound rules** Sharing VPC associations Related resources - new Tags

**Outbound rules (1)** Manage tags Edit outbound rules

Search

| Security group rule ID | IP version | Type        | Protocol | Port range | Destination |
|------------------------|------------|-------------|----------|------------|-------------|
| sgr-02b854597462bf43a  | IPv4       | All traffic | All      | All        | 0.0.0.0/0   |

## 9- create ec2(bastion) in public subnet with security group from 7

Instances (1/1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

< 1 >

| <input checked="" type="checkbox"/> | Name    | Instance ID        | Instance state | Instance type | Status check | Alarm status  | Availability Zone | Public IPv4 DNS |
|-------------------------------------|---------|--------------------|----------------|---------------|--------------|---------------|-------------------|-----------------|
| <input checked="" type="checkbox"/> | bastion | i-Of4777c5177e8eff | Running        | t3.micro      | Initializing | View alarms + | eu-north-1a       | -               |

i-Of4777c5177e8eff (bastion)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM role

-

Owner ID

707938860152

Launch time

Sun May 03 2026 04:58:06 GMT+0300 (Eastern European Summer Time)

Security groups

sg-0d34856d81c2b1e71 (bastion-security-group)

▼ Inbound rules

Filter rules

< 1 >

i-Of4777c5177e8eff (bastion)

Filter rules

< 1 >

| Name | Security group rule ID | Port range | Protocol | Source    | Security groups        |
|------|------------------------|------------|----------|-----------|------------------------|
| -    | sgr-0d9807c072f5dbf40  | 22         | TCP      | 0.0.0.0/0 | bastion-security-group |

▼ Outbound rules

Filter rules

< 1 >

| Name | Security group rule ID | Port range | Protocol | Destination | Security groups        |
|------|------------------------|------------|----------|-------------|------------------------|
| -    | sgr-02c5af1a2f110f62f  | All        | All      | 0.0.0.0/0   | bastion-security-group |

Instances (1/1) Info

Last updated 1 minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

< 1 >

| <input checked="" type="checkbox"/> | Name    | Instance ID        | Instance state | Instance type | Status check | Alarm status  | Availability Zone | Public IPv4 DNS |
|-------------------------------------|---------|--------------------|----------------|---------------|--------------|---------------|-------------------|-----------------|
| <input checked="" type="checkbox"/> | bastion | i-Of4777c5177e8eff | Running        | t3.micro      | Initializing | View alarms + | eu-north-1a       | -               |

i-Of4777c5177e8eff (bastion)

IAM role

-

IMDSv2

Required

Operator

-

▼ Instance details Info

AMI ID

ami-09e320d375d7b8d3e

AMI name

al2023-ami-2023.11.20260427.1-kernel-6.1-x86\_64

Subnet ID

subnet-058e73493146a075d (Public Subnet 1)

Instance ARN

arn:aws:ec2:eu-north-1:707938860152:instance/i-Of4777c5177e8eff

Monitoring

disabled

Allowed image

-

Auto Scaling Group name

-

Managed

false

Platform details

Linux/UNIX

Termination protection

Disabled

# 10- create ec2(application) private subnet with security group from 8

Instances (1/3) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

|                                     | Name        | Instance ID         | Instance state | Instance type | Status check      | Alarm status  | Availability Zone | Public IPv4 DNS |
|-------------------------------------|-------------|---------------------|----------------|---------------|-------------------|---------------|-------------------|-----------------|
| <input checked="" type="checkbox"/> | application | i-09840a1bf23719bc9 | Running        | t3.micro      | 3/3 checks passed | View alarms + | eu-north-1b       | -               |
| <input type="checkbox"/>            | bastion     | i-0f47777c5177e8eff | Terminated     | t3.micro      | -                 | View alarms + | eu-north-1a       | -               |
| <input type="checkbox"/>            | bastion     | i-05c1786ee25251339 | Running        | t3.micro      | Initializing      | View alarms + | eu-north-1a       | -               |

i-09840a1bf23719bc9 (application)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM role

-

Owner ID

707938860152

Launch time

Sun May 03 2026 05:03:20 GMT+0300 (Eastern European Summer Time)

Security groups

sg-07442679122f228f6 (application-security-group)

▼ Inbound rules

Filter rules

< 1 >

Instances (1/3) Info

Last updated 1 minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

|                                     | Name        | Instance ID         | Instance state | Instance type | Status check      | Alarm status  | Availability Zone | Public IPv4 DNS |
|-------------------------------------|-------------|---------------------|----------------|---------------|-------------------|---------------|-------------------|-----------------|
| <input checked="" type="checkbox"/> | application | i-09840a1bf23719bc9 | Running        | t3.micro      | 3/3 checks passed | View alarms + | eu-north-1b       | -               |
| <input type="checkbox"/>            | bastion     | i-0f47777c5177e8eff | Terminated     | t3.micro      | -                 | View alarms + | eu-north-1a       | -               |
| <input type="checkbox"/>            | bastion     | i-05c1786ee25251339 | Running        | t3.micro      | Initializing      | View alarms + | eu-north-1a       | -               |

i-09840a1bf23719bc9 (application)

Filter rules

< 1 >

| Name | Security group rule ID | Port range | Protocol | Source      | Security groups            |
|------|------------------------|------------|----------|-------------|----------------------------|
| -    | sgr-0d728783fc4988d02  | 3000       | TCP      | 10.0.0.0/16 | application-security-group |
| -    | sgr-005c55b05542c368a  | 22         | TCP      | 10.0.0.0/16 | application-security-group |

▼ Outbound rules

Filter rules

< 1 >

| Name | Security group rule ID | Port range | Protocol | Destination | Security groups            |
|------|------------------------|------------|----------|-------------|----------------------------|
| -    | sgr-02b854597462bf43a  | All        | All      | 0.0.0.0/0   | application-security-group |

|                                                                                          |                                                                                                                                 |                                                                                                                                                        |
|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>i-09840a1bf23719bc9 (application)</div><div><div></div><div></div></div></div> |                                                                                                                                 |                                                                                                                                                        |
| <div><div></div><div></div></div>                                                        | <div><div></div><div>vpc-0a0af0e3a076d82a9 (main-VPC) <a href="#">↗</a></div></div>                                             | <div><div></div><div><a href="#">ⓘ Opt-in to AWS Compute Optimizer for recommendations.</a>   <a href="#">Learn more</a> <a href="#">↗</a></div></div> |
| <div><div>IAM role</div><div>-</div></div>                                               | <div><div>Subnet ID</div><div><a href="#">↗</a> subnet-05cd9dd7337b46e43 (Private Subnet 1) <a href="#">↗</a></div></div>       | <div><div>Auto Scaling Group name</div><div>-</div></div>                                                                                              |
| <div><div>IMDSv2</div><div>Required</div></div>                                          | <div><div>Instance ARN</div><div><a href="#">↗</a> arn:aws:ec2:eu-north-1:707938860152:instance/i-09840a1bf23719bc9</div></div> | <div><div>Managed</div><div>false</div></div>                                                                                                          |
| <div><div>Operator</div><div>-</div></div>                                               |                                                                                                                                 |                                                                                                                                                        |
| <div><div>▼ Instance details <a href="#">Info</a></div></div>                            |                                                                                                                                 |                                                                                                                                                        |
| <div><div>AMI ID</div><div><a href="#">↗</a> ami-09e320d375d7b8d3e</div></div>           | <div><div>Monitoring</div><div>disabled</div></div>                                                                             | <div><div>Platform details</div><div><a href="#">↗</a> Linux/UNIX</div></div>                                                                          |